

# TIP SHEET: UNDERSTANDING DATA ECOSYSTEMS

OCHA CENTRE FOR HUMANITARIAN DATA

## INTRODUCTION

A data ecosystem consists of the data management activities and related data, infrastructure, tools, processes, stakeholders and applicable guidance in a context. Understanding how the different components of a data ecosystem relate to one another and how data moves within and between organizations helps to ensure that data is managed in a responsible way.

There are three steps to understanding data ecosystems in a humanitarian response:

- **Developing an organization-level data management registry and activity diagrams.**
- **Compiling registries at the cluster/sector-level and system-wide level.**
- **Developing data ecosystem maps.**

This tip sheet offers instructions and templates for completing these steps in line with the IASC Operational Guidance on Data Responsibility in Humanitarian Action.<sup>1</sup>

## STEP 1: DEVELOPING AN ORGANIZATION-LEVEL DATA MANAGEMENT REGISTRY AND ACTIVITY DIAGRAMS

The first step to understanding a data ecosystem is documenting each of the data management activities led by your organization in the context. Examples of data management activities include needs assessments, displacement tracking, beneficiary registration and enrollment, protection case management, community perception surveys, and response monitoring and evaluation, among others.

Organizations should document these activities in a **data management registry**, which typically includes the following details:

- **Activity description**, including type (e.g. registration, needs assessment, survey), purpose, location and sensitivity of the data being managed.
- **Timeframe and status**, including whether the activity is one-off or recurring, and whether it is ongoing, completed or planned.
- **Actors and responsibilities**, including the lead actor, collaborators, roles, and relevant cluster and sub-cluster/working group/area of responsibility.
- **Infrastructure and tools** used for collecting, receiving, storing, analyzing and destroying the data.
- **Data disaggregation**, such as by admin level, sex, age or disability, and if relevant, the methodology used.
- **Data sharing**, including the websites, reports and information products the data management activity will contribute to.
- **Guidance and governance**, including Standard Operating Procedures for the activity, applicable legislation and regulations, and applicable Information Sharing Protocols or similar documents.

<sup>1</sup> For reference see the [IASC Operational Guidance on Data Responsibility in Humanitarian Action](#).

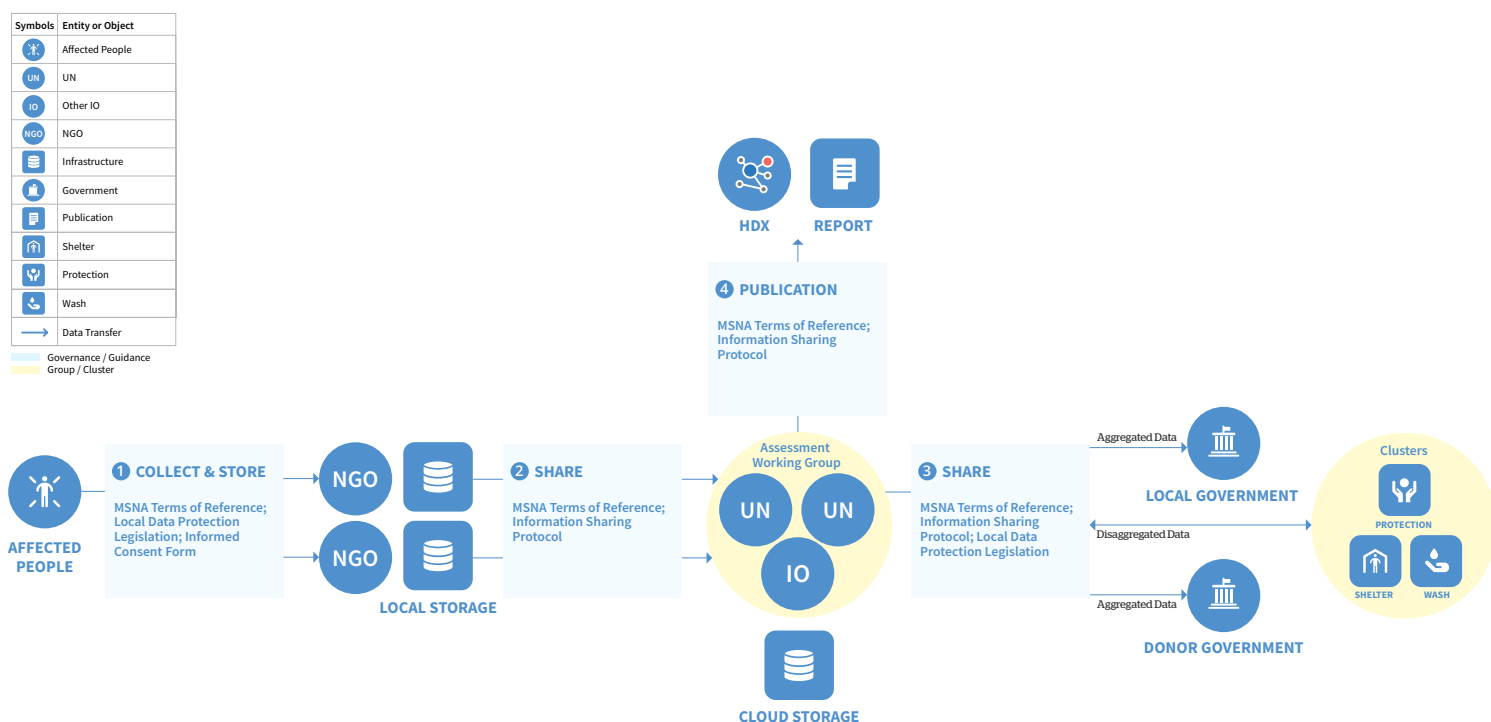
Use the [Data Management Registry Template](#) to develop and maintain the registry for your organization.

The details of each data management activity can also be visualized in a **data management diagram**. A diagram offers a visual understanding of an activity and can help identify gaps in data sharing, infrastructure or guidance. Use the [Data Management Diagram Template](#) to develop the diagram. The organization-level registry and diagrams serve as the basis for step two described below.

The data management diagram below shows the steps of a hypothetical Multi-Sector Needs Assessment (MSNA).

1. Two NGOs collect data from affected people after obtaining consent using the MSNA Informed Consent Form. The NGOs store the data locally on servers in their offices in line with the MSNA Terms of Reference (ToR) and local data protection legislation.
2. The NGOs share the data with the Assessment Working Group (AWG), which consists of two UN entities and an International Organization. The AWG compiles the data and stores it in the cloud.
3. The AWG shares disaggregated data with the clusters for analysis, which the clusters share back with the AWG once complete. The AWG also shares aggregated data with the local government and a donor government in line with the Information Sharing Protocol (ISP).
4. The AWG synthesizes the clusters' analysis and publishes a needs assessment report. It also shares the underlying aggregated data on the Humanitarian Data Exchange.

## DATA MANAGEMENT DIAGRAM: A MULTI-SECTOR NEEDS ASSESSMENT



## STEP 2: COMPILING REGISTRIES AT THE CLUSTER/SECTOR-LEVEL AND SYSTEM-WIDE LEVEL

The second step to understanding data ecosystems involves sharing the organization-level data management registry and diagrams with relevant clusters/sectors and with the Information Management Working Group (IMWG) or similar coordination group for compilation. This helps identify data gaps and duplication at the cluster/sector-level and system-wide level. It also enables prioritization and strategic decision-making on responsible data management.

Clusters/sector leads should maintain a cluster/sector-level registry of the data management activities led by their members and relevant to their thematic area of focus. For example, a health cluster data management registry might include activities such as disease monitoring, health facility tracking, distribution of medical supplies. The cluster/sector-level registry should be developed based on the [Data Management Registry Template](#).

At the system-wide level, a registry should contain all data management activities in a response, categorized by lead organization and cluster. In most cases, the registry would be compiled by OCHA on behalf of the IMWG, again using the [Data Management Registry Template](#).

Registries should be shared publicly to ensure transparency around data management activities.<sup>2</sup> In some response environments — notably in conflict settings — the fact that a given data management activity is taking place or specific details about the activity may need to be kept confidential. Any risk associated with sharing the registry should be considered by the lead organization for the activity and in consultation with the relevant cluster/sector partners and OCHA.

## STEP 3: DEVELOPING DATA ECOSYSTEM MAPS

The third step is developing data ecosystem maps. A data ecosystem map is a visualization of the various ways that data flows between and among actors in multiple data management activities. OCHA should develop a data ecosystem map at the system-wide level, in consultation with the IMWG and ICCG.

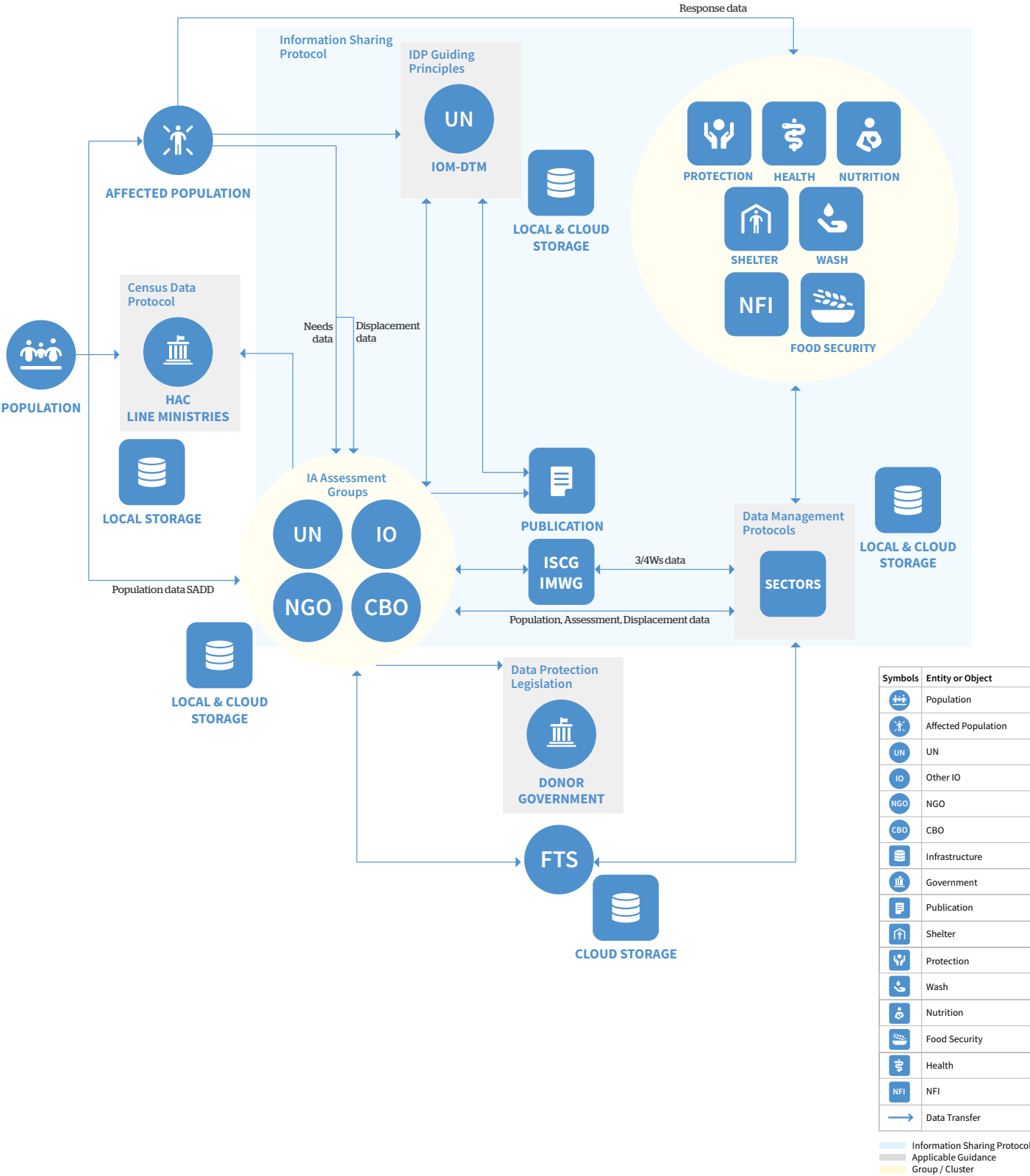
Given the complexity of data ecosystems, it may be best to use a white-board or flip chart to sketch out the priority data management activities in your response. If useful, you can then create a digital version using the visualization tools and software designed for this purpose.<sup>3</sup> Some details may be excluded to reduce complexity and improve the legibility of the data ecosystem map.

The data ecosystem map below represents a data ecosystem in which four hypothetical data management activities are taking place. These include a mobility tracking activity, a needs assessment, internal displacement tracking, and response monitoring. Visualizing these data management activities together helps gain a quick understanding of the actors involved in data management across the response, how data flows between those actors and where there is room for improving the data ecosystem.

<sup>2</sup> For example [OCHA's 2021 Assessment Registry for Palestine](#).

<sup>3</sup> Tools include the [Open Data Institute Data Ecosystem Mapping tool](#), the [Protection Information Management Training Resource Pack](#) (see package 4 from page 163) or the template provided in Annex D of the [IASC Operational Guidance on Data Responsibility in Humanitarian Action](#). [Microsoft Visio](#) is a suitable tool to visualize data ecosystem maps. Free online tools should be used with caution given data security risks.

# POPULATION, ASSESSMENT, INTERNAL DISPLACEMENT & RESPONSE MONITORING ECOSYSTEM MAP



Data ecosystem maps should be completed and updated annually by OCHA in consultation with the IMWG and ICCG. They can be shared publicly unless they contain data management activities that should be kept confidential. They should also be shared with the Humanitarian Country Team for reference.

## CONCLUSION

The development of data management registries, diagrams and data ecosystem maps improves the transparency of data management in humanitarian response. Following the three steps presented in this tip sheet can lead to a better understanding of the actors, data flows and opportunities for improving data management across a data ecosystem. The OCHA Centre for Humanitarian Data is available for advisory support to offices and organizations interested in developing these outputs.

COLLABORATORS: JOINT IDP PROFILING SERVICE (JIPS).

The **Centre for Humanitarian Data** ('the Center'), together with key partners, is publishing a series of guidance notes and tip sheets on Data Responsibility in Humanitarian Action over the course of 2022 and 2023. The guidance note series continues work initiated in 2019 and 2020. It also complements the **Inter-Agency Standing Committee Operational Guidance on Data Responsibility in Humanitarian Action** and the **OCHA Data Responsibility Guidelines**, which were published in February 2021 and October 2021 respectively. Through the series, the Centre aims to provide additional guidance on specific issues, processes and tools for data responsibility in practice. The guidance notes and tip sheets published in 2022-2023 have been made possible with the support of the Government of Switzerland.



Schweizerische Eidgenossenschaft  
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Federal Department of Foreign Affairs FDFA

STATE OF WISCONSIN

PERSONNEL COMMISSION

\* \* \* \* \*  
KARL W. MARQUARDT,  
Appellant,  
v.  
Superintendent, DEPARTMENT OF  
PUBLIC INSTRUCTION,  
Respondent.  
Case No. 90-0349-PC  
\* \* \* \* \*

RULING ON  
MOTION  
TO DISMISS

This matter is before the Commission following a prehearing conference, November 5, 1990, in which respondent raised the issue of the Commission's authority to hear the captioned appeal. Subsequently, on November 19, 1990, respondent filed a motion to dismiss the captioned appeal because it was untimely.

For purposes of this motion the following facts are taken as uncontested.

On March 15, 1990, respondent announced a servicewide promotional opportunity for its chief legal counsel position in the Department of Employment Relation's Job Opportunities Bulletin.

Appellant applied for the position, went through the examination process and was among those certified as eligible for appointment. Five candidates were interviewed during the final screening which was completed on or about May 30, 1990.

On July 3, 1990, appellant received notice that he was not selected for the position. Subsequently, about July 8, appellant learned the position may have been filled by a transfer candidate not on the certification list.

Appellant requested respondent to provide a copy of the certification list and identification of the appointee. He received it on July 18, 1990. Respondent's cover letter indicated July 15, 1990, was the effective dates of the appointee's appointment.

On August 14, 1990, appellant filed an appeal with the Commission, alleging respondent violated §230.44(1)(d), Stats.

Respondent contends the Commission lacks jurisdiction over this matter because appellant failed to file his appeal within thirty days after he was notified of his nonselection. Respondent cites Cozzens-Ellis v. Personnel Comm., 155 Wis 2d 271, 455 N.W. 2d 246 (Ct. App. 1990), which held the effective date of promotion and not the date the person who was promoted began the new job was the decision date, was on point.

In response, appellant argues that Cozzens-Ellis is distinguishable from this matter on the basis of facts and on the basis of the subject of the appeal. Appellant alleges he is appealing the hiring process and not his non-selection.

The facts in Cozzens-Ellis and this matter are virtually identical. They differ only in the breadth of the job promotional opportunity where in this matter the promotional opportunity was servicewide, while in Cozzens-Ellis it was intra-departmental. This distinction has little bearing on the matter.

Concerning appellant's second argument, he alleges he seeks a review of: "the hiring process, i.e., changing methods for obtaining a pool of applicants without adequate notice to afford an equal opportunity for all interested candidates to be considered under the transfer opportunity, . . ."

In Cozzens-Ellis the court said:

Under Sec. 230.44(1)(d), an employee appeals from a 'personnel action' alleged to be illegal or an abuse of discretion. If a person is denied a promotion, the action appealed from is the denial, not a later event stemming from it.

In the instant case before the Commission, appellant, as in Cozzens-Ellis, appealed under §230.44(1)(d). Consequently, the action he appealed was his denial or nonselection. However, if, as appellant alleges, §230.44(1)(d) includes 'a general appeal of the hiring process, then the focus of appellant's appeal is respondent's "action" allowing the inclusion of transfer employees in the pool of eligible candidates and not a later event stemming from it. In either instance, undisputed facts show that appellant filed his appeal more than thirty days after the effective date of respondent's personnel action allowing the inclusion of transferees or his notice of nonselection.

For the above stated reasons, respondent's motion for dismissal prevails.

ORDER

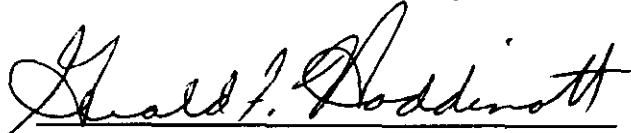
This appeal is dismissed as untimely filed.

Dated: January 11, 1991      STATE PERSONNEL COMMISSION

  
LAURIE R. McCALLUM, Chairperson

DRM/gdt/2

  
DONALD R. MURPHY, Commissioner

  
GERALD F. HODDINOTT, Commissioner

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U.S. Congressman Mike Pence will speak at Delta High School's second "Energy Summit" on Friday, September 10. The Congressman will speak to around 400 high school students about energy policy.

Friday, September 10, 2010:

**11:30 AM** – Congressman Pence to speak at Delta High School "Energy Summit"

Delta High School  
3400 SR 28  
Muncie

*April 28, 2009*

**Mr. SCOTT of Virginia:** Madam Speaker, the gentleman from Massachusetts mentioned the fact that, over the last few years, we'd gotten ourselves into the ditch. This shows the ditch that we're actually in.

In 1993, we passed a budget that dug ourselves out of a ditch and created surpluses, as far as I could see. In fact, in 2001, when we came into session, we had a surplus sufficient to put us on track to paying off the entire national debt held by the public by last year. Instead, we had a complete collapse of the budget beginning in 2001, and there is no telling where this line is going to end up. It took 8 years to get into this ditch.

During the good years when we had fiscal responsibility, not only were we on the way to paying off the national debt, but we created record numbers of jobs. We had a median income increase of about \$7,000 per family, and the Dow Jones Industrial Average more than tripled. Now we have a situation where we have had the worst job performance since the Great Depression, where the median income is actually down when adjusted for inflation and where the Dow Jones Industrial Average is worse than it was when it started. It took us 8 years to get into this ditch.

We have an urgent situation. This budget will cut the deficit in half in 4 years. Now, that is not the end of it. That's not enough. Cutting the deficit in half is not enough, but for one year's work, that is certainly a good step toward getting us out of a ditch that took 8 years to get us into.

Now we have a situation where the new budget will restore PAYGO, that is, that any new program will have to be paid for. The reason we could get it in this kind of ditch was we passed tax cuts that we hadn't paid for, and we had spending that wasn't paid for. But under this budget, any new initiative will have to be paid for, and that's going to be hard. We're talking about energy initiatives. We're talking about health care initiatives and education initiatives that will be very expensive, but none of them can go into effect unless they're paid for with other

spending cuts or with tax increases. Everything will be paid for. This is in stark contrast to what happened in 2001 when we didn't pay for anything. We went right into a ditch, and we didn't create any jobs.

It is urgent that we pass this budget to get back on the track that we were on in 1993 when the budget created jobs, when the median income was up, when the economy was good, and when we were on the way to paying off the national debt, instead of the ditch we're in today where we have had, in the last 8 years, the worst job performance since the Great Depression and huge deficits as far as the eye can see. We're taking a major step in the right direction.

So, Madam Speaker, I would hope that we would adopt the budget so we could get on to the job of restoring the economy and of balancing this budget.